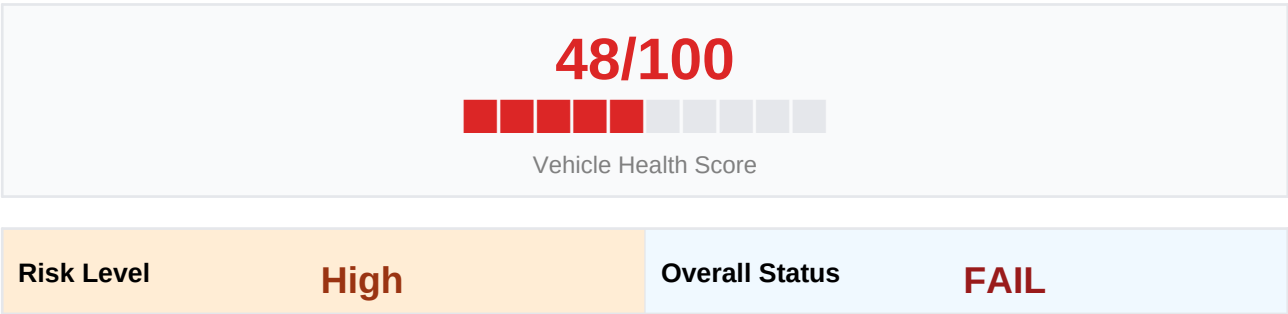


# VEHICLE INSPECTION REPORT

AI-Powered Body + Engine Sound Analysis · June 07, 2026 at 16:42  
Physical Defect Detection (7-class YOLO + Severity Model) + Engine Knock Audio Classification

|                    |                                            |         |              |
|--------------------|--------------------------------------------|---------|--------------|
| VIN / Registration | WDDZF8DBXRA765432                          | Mileage | Not Provided |
| Make / Model       | MERCEDES-BENZ MERCEDES-BE<br>NZ-SEDAN-TEST | Year    | 2024         |

## AI Vehicle Health Assessment



### Key Risk Factors:

- Safety-critical components affected: Light — Lamp Crack, Glass Crack
- High-confidence damage on: Light — Lamp Crack, Fender — Scratch, Scratch
- Minor cosmetic damage on: Door — Scratch, Part Crack

### AI Inspection Summary

The inspection detected: Light — Lamp Crack, Door — Scratch, Fender — Scratch, Scratch, Part Crack, Glass Crack. Safety-critical damage identified on Light — Lamp Crack, Glass Crack requires immediate attention.

### Defect Analysis — Dual Model Results

- Light — Lamp Crack

■ Safety-Critical

(Severe, 80.2% confidence)

Damaged lights reduce road legality — replace before driving at night.
- Door — Scratch

(Moderate, 65.1% confidence)

Inspect door seals and hinges — repair to prevent water ingress.
- Fender — Scratch

(Severe, 88.7% confidence)

Exposed metal will oxidise — treat and repair at nearest opportunity.
- Door — Scratch

(Minor, 48.7% confidence)

Inspect door seals and hinges — repair to prevent water ingress.
- Scratch

(Moderate, 76.8% confidence)

Scratches expose bare metal — treat with touch-up paint to prevent rust.

**Part Crack** (Moderate, 74.8% confidence)  
Consult a qualified technician for assessment and repair.

**Part Crack** (Moderate, 69.8% confidence)  
Consult a qualified technician for assessment and repair.

**Glass Crack ■ Safety-Critical** (Moderate, 60.1% confidence)  
Glass cracks compromise structural integrity and visibility — replace immediately.

**AI Recommendations**

- 1. Damaged lights reduce road legality — replace before driving at night.
- 2. Inspect door seals and hinges — repair to prevent water ingress.
- 3. Exposed metal will oxidise — treat and repair at nearest opportunity.
- 4. Scratches expose bare metal — treat with touch-up paint to prevent rust.
- 5. Cracked component detected — assess structural impact and repair.

**Standard 7-Part Damage Checklist**

| Component                | Defect Type / Severity | Confidence      | Status   |
|--------------------------|------------------------|-----------------|----------|
| Bonnet                   | No Damage Detected     | —               | —        |
| Bumper                   | No Damage Detected     | —               | —        |
| Dickey                   | No Damage Detected     | —               | —        |
| Door                     | Door — Scratch         | 65.1%           | Moderate |
| Fender                   | Fender — Scratch       | 88.7%           | Severe   |
| Light                    | Light — Lamp Crack     | 80.2%           | Severe ■ |
| Windshield               | No Damage Detected     | —               | —        |
| Total Unique Parts/Types | 6                      | Overall Status: | FAIL     |

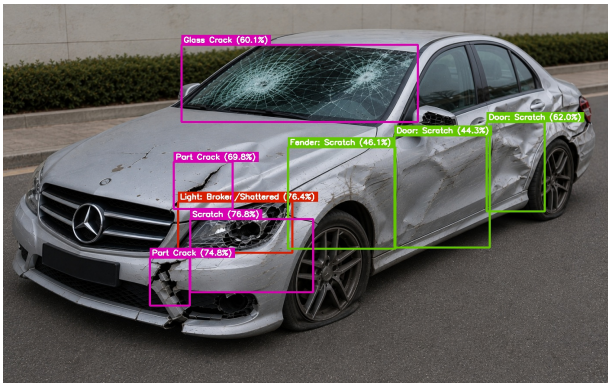
**Additional Damage Detections (Severity Model)**

| Damage Type | Confidence | Severity Tier |
|-------------|------------|---------------|
| Scratch     | 76.8%      | Moderate      |
| Part Crack  | 74.8%      | Moderate      |

|             |       |          |
|-------------|-------|----------|
| Part Crack  | 69.8% | Moderate |
| Glass Crack | 60.1% | Moderate |

## Annotated Images — Dual Model Detections

Blue/green boxes = part detections; Purple boxes = severity-only detections



*This report was generated using the unified MEHRA inspection pipeline: physical defect detection with a 7-class part model and severity model, engine sound knock classification, and Groq LLaMA 3.3 for report analysis. Results should be verified by a qualified technician.*